The Garland–Lepowsky theorem in the chiral setting: bar cohomology and Koszul duality for affine Kac–Moody algebras

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Abstract

The classical Garland–Lepowsky theorem computes the Lie algebra cohomology $H^p(\mathfrak{n}_-;\mathbb{C})$ of the negative nilpotent subalgebra of a Kac–Moody algebra in terms of Weyl group combinatorics. We prove a chiral extension: for the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, the bar cohomology $H^n(\bar{B}(V_k(\mathfrak{g})))$ is concentrated in bar degree 1 for each conformal weight, with $\dim H^n = (2n+1)\dim\mathfrak{g}/3$ for $\mathfrak{g} = \mathfrak{sl}_2$ and, more generally, polynomial growth governed by the Weyl group of $\mathfrak{g}$. The concentration is equivalent to chiral Koszulness of $V_k(\mathfrak{g})$ via the PBW spectral sequence. The Koszul dual is the chiral Chevalley–Eilenberg algebra at the Feigin–Frenkel dual level $k' = -k - 2h^\vee$, with modular characteristic $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ satisfying $\kappa + \kappa' = 0$. The shadow obstruction tower is class L with depth $r_{\max} = 3$: a nonzero cubic shadow from the Lie bracket, killed at the quartic level by the Jacobi identity.

1 Introduction

Kostant's 1961 theorem [7] computes the Lie algebra cohomology of the nilpotent radical of a Borel subalgebra of a finite-dimensional semisimple Lie algebra: the cohomology $H^p(\mathfrak{n}_-;\mathbb{C})$ is concentrated in a single Weyl group orbit at each cohomological degree p, with dimension equal to the number of Weyl group elements of length p. Garland and Lepowsky [6] extended this to the infinite-dimensional setting of Kac–Moody algebras, where the negative subalgebra $\mathfrak{n}_- = \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$ is infinite-dimensional and the cohomology acquires a grading by conformal weight.

The present paper proves a further extension: the chiral bar complex $\bar{B}(V_k(\mathfrak{g}))$ of the universal affine vertex algebra at non-critical level has bar cohomology concentrated in bar degree 1 at each conformal weight. This concentration is the content of the chiral Koszulness of $V_k(\mathfrak{g})$, one of the twelve equivalent characterizations proved in the monograph [9]. The mechanism is the PBW spectral sequence: the E_1 page is the Chevalley–Eilenberg complex of the negative loop algebra $\mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$ with coefficients in the trivial module, and the classical Garland–Lepowsky concentration forces collapse at E_2.

The paper is organized as follows. Section 2 recalls the classical Garland–Lepowsky theorem and Kostant's theorem for finite-dimensional $\mathfrak{g}$. Section 3 constructs the chiral bar complex of $V_k(\mathfrak{g})$ and identifies its differential with the Chevalley–Eilenberg differential of the loop algebra. Section 4 proves chiral Koszulness via PBW spectral sequence collapse. Section 5 identifies the Koszul dual algebra and the Feigin–Frenkel involution. Section 6 computes the shadow depth classification. Section 7 works out the explicit cases $\mathfrak{sl}_2$, $\mathfrak{sl}_3$, and the exceptional types.

2 The classical Garland–Lepowsky theorem

2.1 Kostant's theorem for finite-dimensional $\mathfrak{g}$

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root system Δ, positive roots Δ_+, Weyl group W, and Borel subalgebra $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$ where $\mathfrak{n}_+ = \bigoplus_{\alpha\in\Delta_+} \mathfrak{g}_\alpha$. Let $\mathfrak{n}_- = \bigoplus_{\alpha\in\Delta_+} \mathfrak{g}_{-\alpha}$ be the opposite nilpotent subalgebra, and let ρ denote the Weyl vector (half the sum of positive roots).

Theorem 2.1 (Kostant [7])**.** *For each $p \geq 0$, the Lie algebra cohomology $H^p(\mathfrak{n}_-;\mathbb{C})$ is a semisimple $\mathfrak{h}$-module:*

$$H^p(\mathfrak{n}_-;\mathbb{C}) \;\cong\; \bigoplus_{\substack{w\in W\\ \ell(w)=p}} \mathbb{C}_{w\cdot 0}, \tag{2.1}$$

where $w\cdot 0 = w(\rho)-\rho$ is the dot action and $\ell(w)$ is the length of w. In particular, $\dim H^p(\mathfrak{n}_-;\mathbb{C}) = |\{w \in W : \ell(w) = p\}|$.

For $\mathfrak{g} = \mathfrak{sl}_2$: $W = \{1, s\}$, $\ell(1) = 0$, $\ell(s) = 1$, so $\dim H^0 = 1$, $\dim H^1 = 1$, $H^p = 0$ for $p \geq 2$. For $\mathfrak{g} = \mathfrak{sl}_3$: $|W| = 6$, with length distribution $(1, 2, 2, 1)$, giving $\dim H^p = (1, 2, 2, 1)$ for $p = 0, 1, 2, 3$.

More generally, for a highest-weight module V_λ with highest weight λ:

Theorem 2.2 (Kostant [7])**.** *$H^p(\mathfrak{n}_-; V_\lambda) \cong \bigoplus_{\ell(w)=p} V^{\mathfrak{h}}_{w\cdot\lambda}$, where the sum runs over $w \in W$ with $\ell(w) = p$ such that $w \cdot \lambda$ is dominant.*

The proof uses the Hodge decomposition for the Laplacian $\Box = dd^* + d^*d$ on $\Lambda^*(\mathfrak{n}_-^*)$, where the adjoint d^* is defined using the Killing form. The key input is that $\mathfrak{g}$ is semisimple: the Casimir operator provides the Hodge-theoretic splitting.

2.2 Extension to Kac–Moody algebras

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, $\widehat{\mathfrak{g}} = \mathfrak{g} \otimes \mathbb{C}((t)) \oplus \mathbb{C}K$ the associated affine Kac–Moody algebra, and

$$\mathfrak{n}_-^{\mathrm{loop}} \;=\; \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}] \tag{2.2}$$

the negative part of the loop algebra (the subalgebra spanned by J^a_{-m} for $m \geq 1$ and $a = 1, \ldots, \dim \mathfrak{g}$). This is a graded Lie algebra with the conformal weight grading $|J^a_{-m}| = m$.

Theorem 2.3 (Garland–Lepowsky [6])**.** *The Lie algebra cohomology $H^p(\mathfrak{n}_-^{\mathrm{loop}};\mathbb{C})$ is finite-dimensional in each conformal weight h and satisfies:*

(i) At each p and h, the cohomology is concentrated in a single bidegree: for each cohomological degree p, H^p is nonzero at only finitely many conformal weights, and the nonzero weights are determined by the affine Weyl group W_{aff}.

(ii) The Euler characteristic of H^* at conformal weight h equals the coefficient of q^h in the Euler series

$$\chi(\bar{B}(V_k(\mathfrak{g}))) \;=\; \prod_{n\geq 1}(1-q^n)^{\dim \mathfrak{g}}. \tag{2.3}$$

(iii) For semisimple $\mathfrak{g}$, the cohomology H^p at each conformal weight is concentrated in a single cohomological degree (Kostant-type concentration), so the Euler characteristic equals the total dimension up to sign.

Remark 2.4. The mechanism of concentration is the semisimplicity of $\mathfrak{g}$. The Chevalley–Eilenberg complex $C^*(\mathfrak{n}_-^{\mathrm{loop}};\mathbb{C}) = \Lambda^*(\mathfrak{n}_-^{\mathrm{loop}})^*$ admits a PBW-type filtration by the number of distinct negative modes. The E_1 page of the associated spectral sequence computes $H^*(\mathfrak{g};M)$ for $\mathfrak{g}$-modules M arising as graded components of the symmetric algebra on the loop generators. The Whitehead lemma — $H^p(\mathfrak{g};M)=0$ for $p\geq 1$ and M a nontrivial finite-dimensional simple $\mathfrak{g}$-module — kills all terms except the $\mathfrak{g}$-invariant part. The surviving terms are computed by Kostant's theorem.

3 The chiral bar complex of $V_k(\mathfrak{g})$

3.1 The universal affine vertex algebra

Let $\mathfrak{g}$ be a simple Lie algebra with invariant bilinear form $(\cdot,\cdot)$ normalized so that the highest root θ satisfies $(\theta,\theta)=2$, and let $h^\vee$ denote the dual Coxeter number. The universal affine vertex algebra $V_k(\mathfrak{g})$ at level k is freely strongly generated by the currents $\{J^a(z)\}_{a=1}^{\dim\mathfrak{g}}$ with OPE

$$J^a(z)\,J^b(w) \;\sim\; \frac{k\,(J^a,J^b)}{(z-w)^2} \;+\; \frac{f^{ab}{}_c\,J^c(w)}{z-w}, \tag{3.1}$$

where $f^{ab}{}_c$ are the structure constants of $\mathfrak{g}$ in the orthonormal basis. Free strong generation means the normally ordered monomials $:\partial^{n_1}J^{a_1}\cdots\partial^{n_r}J^{a_r}:$ form a PBW basis of $V_k(\mathfrak{g})$.

3.2 Construction of the bar complex

The ordered bar complex of $V_k(\mathfrak{g})$ is the cofree tensor coalgebra

$$B^{\mathrm{ord}}(V_k(\mathfrak{g})) \;=\; T^c(s^{-1}\overline{V}_k(\mathfrak{g})) \;=\; \bigoplus_{n\geq 0}(s^{-1}\overline{V}_k(\mathfrak{g}))^{\otimes n}, \tag{3.2}$$

where $\overline{V}_k(\mathfrak{g}) = V_k(\mathfrak{g})/\mathbb{C}|0\rangle$ is the augmentation ideal and s^{-1} denotes desuspension (lowering cohomological degree by 1). The bar differential d_B is the unique coderivation whose Taylor components $d_B^{(n)}\colon (s^{-1}\overline{V})^{\otimes n}\to s^{-1}\overline{V}$ encode the chiral algebra operations: the binary component $d_B^{(2)}$ extracts the OPE via the logarithmic kernel $d\log(z_1-z_2)$, and the nullary component $d_B^{(0)}$ records the curvature.

Geometrically, the degree-n piece of the bar complex is

$$\bar{B}^n(V_k(\mathfrak{g})) \;=\; \Gamma\Big(\overline{C}_{n+1}(X),\; \mathfrak{g}^{\boxtimes(n+1)}\otimes\omega_X^{\boxtimes(n+1)}\otimes\Omega^n_{\log}\Big), \tag{3.3}$$

where $\overline{C}_{n+1}(X)$ is the Fulton–MacPherson compactification [5] of the configuration space of $n+1$ points on a smooth curve X, and $\Omega^n_{\log}$ is the sheaf of logarithmic n-forms along the boundary divisors.

Proposition 3.1. *The bar differential on degree-1 elements $J^a\boxtimes J^b\cdot\eta_{12}$, where $\eta_{12}=d\log(z_1-z_2)$, is:*

$$d_B(J^a\boxtimes J^b\cdot\eta_{12}) \;=\; k\,(J^a,J^b)\cdot|0\rangle \;+\; f^{ab}{}_c\,J^c. \tag{3.4}$$

The first term arises from the double pole in (3.1) (the $d\log$ kernel absorbs one power of $(z-w)$, converting a double pole to a residue); the second from the simple pole.

Proof. The residue extraction $\mathrm{Res}_{z_1=z_2}[J^a(z_1)J^b(z_2)\cdot d\log(z_1-z_2)]$ applies $d\log(z_1-z_2) = dz_1/(z_1-z_2)$ to the OPE

$$J^a(z_1)J^b(z_2) \;=\; \frac{k\,(J^a,J^b)}{(z_1-z_2)^2} + \frac{f^{ab}{}_c\,J^c(z_2)}{z_1-z_2} + \text{regular}.$$

The $d\log$ absorption converts the double pole to a simple pole with residue $k\,(J^a,J^b)$, and the simple pole to a constant with value $f^{ab}{}_c\,J^c(z_2)$. □

3.3 Curvature of the bar complex

Proposition 3.2. *The bar differential satisfies $d_B^2 = 0$ as a coderivation on $T^c(s^{-1}\overline{V})$. The A_∞-curvature m_0 measures the failure of $m_1^2 = 0$ on the generating space:*

$$m_1^2 \;=\; \frac{k+h^\vee}{2h^\vee}\cdot\kappa_{\mathfrak{g}}, \tag{3.5}$$

where $\kappa_{\mathfrak{g}} = \sum_a J^a\otimes J_a$ is the Casimir element. The factor $h^\vee$ arises from the identity $\sum_{c,d} f^{ac}{}_d\,f^{bc}{}_d = 2h^\vee\cdot\delta^{ab}$ (the Killing form in the orthonormal basis).

Proof. Acting with d_B twice on $J^a\boxtimes J^b\cdot\eta_{12}$ and applying (3.4):

$$\begin{aligned} d_B^2(J^a\boxtimes J^b\cdot\eta_{12}) &= d_B\big(k\,(J^a,J^b)\cdot|0\rangle + f^{ab}{}_c\,J^c\big)\\ &= f^{ab}{}_c\cdot d_B(J^c)\\ &= f^{ab}{}_c\cdot\big(\text{terms involving } d_B^{(2)}(J^c\boxtimes\cdot)\big). \end{aligned}$$

The Jacobi identity $f^{ab}{}_c\,f^{cd}{}_e + \text{cyclic} = 0$ combines with the Killing form computation $\sum_{c,d} f^{ac}{}_d\,f^{bc}{}_d = 2h^\vee\,\delta^{ab}$ to give the claimed factor $(k+h^\vee)/(2h^\vee)$. At the critical level $k=-h^\vee$, the curvature vanishes and $m_1^2=0$. □

3.4 Identification with Chevalley–Eilenberg cohomology

The PBW filtration on $V_k(\mathfrak{g})$ induces a filtration on the bar complex. The associated graded is the symmetric algebra $\mathrm{Sym}(\mathfrak{g}[t^{-1}]t^{-1})$, and the E_1 page of the resulting spectral sequence is the Chevalley–Eilenberg complex:

Proposition 3.3. *The E_1 page of the PBW spectral sequence on $\bar{B}(V_k(\mathfrak{g}))$ is isomorphic to the Chevalley–Eilenberg cochain complex $C^*_{\mathrm{CE}}(\mathfrak{n}_-^{\mathrm{loop}};\mathbb{C})$ of the negative loop algebra $\mathfrak{n}_-^{\mathrm{loop}} = \mathfrak{g}\otimes t^{-1}\mathbb{C}[t^{-1}]$ with trivial coefficients.*

Proof. The PBW filtration $F_p(V_k(\mathfrak{g}))$ is defined by the total number of generators in a normally ordered monomial. On the associated graded, the OPE reduces to the commutative product (the singular part is lower-order in the filtration), and the bar differential reduces to the Lie bracket component $f^{ab}{}_c$ acting on the exterior algebra $\Lambda^*(\mathfrak{g}[t^{-1}]t^{-1})^*$. This is precisely the Chevalley–Eilenberg differential. □

4 Chiral Koszulness

Definition 4.1. A chiral algebra $\mathcal{A}$ is *chirally Koszul* if the bar cohomology $H^*(\bar{B}(\mathcal{A}))$ is concentrated in bar degree 1 at each conformal weight:

$$H^n(\bar{B}(\mathcal{A}))_h \neq 0 \quad \Longrightarrow \quad n = 1. \tag{4.1}$$

Equivalently, the PBW spectral sequence collapses at E_2 (see [9], Theorem 5.1.1).

Theorem 4.2 (Chiral Koszulness of $V_k(\mathfrak{g})$)**.** *For any simple Lie algebra $\mathfrak{g}$ and non-critical level $k \neq -h^\vee$, the universal affine vertex algebra $V_k(\mathfrak{g})$ is chirally Koszul.*

Proof. The proof has two steps: PBW spectral sequence collapse and free strong generation.

Step 1: PBW spectral sequence collapse. By Proposition 3.3, the E_1 page of the PBW spectral sequence is the Chevalley–Eilenberg complex $C^*_{\mathrm{CE}}(\mathfrak{n}_-^{\mathrm{loop}};\mathbb{C})$. The E_2 page is therefore $H^*_{\mathrm{CE}}(\mathfrak{n}_-^{\mathrm{loop}};\mathbb{C})$, which is computed by the Garland–Lepowsky theorem (Theorem 2.3).

The Garland–Lepowsky concentration (property (iii) of Theorem 2.3) states that for semisimple $\mathfrak{g}$, the cohomology at each conformal weight is concentrated in a single cohomological degree. This concentration is a consequence of the Whitehead vanishing lemma applied to $\mathfrak{g}$-modules appearing in the spectral sequence: for each negative mode J^a_{-m} with $m \geq 1$, the $\mathfrak{g}$-module $\mathfrak{g}_{-m} = \mathfrak{g} \otimes t^{-m}$ is isomorphic to the adjoint representation, which is simple for simple $\mathfrak{g}$. The Whitehead lemma then forces $H^p(\mathfrak{g};\mathfrak{g}_{-m}) = 0$ for $p \geq 1$, and the spectral sequence collapses at E_2.

Step 2: Free strong generation. The universal affine vertex algebra $V_k(\mathfrak{g})$ at any level k is freely strongly generated by the currents $\{J^a\}_{a=1}^{\dim\mathfrak{g}}$: the normally ordered monomials form a PBW basis with no relations (the vacuum Verma module has no null vectors at generic level). By the PBW universality criterion (Proposition 5.3.1 of [9]): any freely strongly generated chiral algebra whose associated graded is classically Koszul is chirally Koszul. Since $\mathrm{gr}_F(V_k(\mathfrak{g})) \cong \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}_k)$, and the symmetric algebra is Koszul by Priddy's theorem [12], the result follows. □

Remark 4.3. The theorem admits two independent proofs. The first, given above, uses the Garland–Lepowsky concentration to force spectral sequence collapse. The second uses the abstract PBW universality criterion, which requires only free strong generation and Koszulness of the associated graded. That both proofs yield the same conclusion is a consistency check. The first proof gives more: it identifies the bar cohomology with Lie algebra cohomology of the loop algebra, providing explicit dimensions.

4.1 The Whitehead mechanism in detail

The spectral sequence collapse in Step 1 of the proof of Theorem 4.2 deserves a closer analysis, since it is the mechanism by which the classical Garland–Lepowsky theorem enters the chiral world.

The negative loop algebra $\mathfrak{n}_-^{\mathrm{loop}} = \mathfrak{g} \otimes t^{-1}\mathbb{C}[t^{-1}]$ admits a conformal weight decomposition $\mathfrak{n}_-^{\mathrm{loop}} = \bigoplus_{m\geq 1} \mathfrak{g}_{-m}$, where $\mathfrak{g}_{-m} = \mathfrak{g} \otimes t^{-m} \cong \mathfrak{g}$ as a $\mathfrak{g}$-module under the adjoint action. The Chevalley–Eilenberg complex decomposes accordingly:

$$C^p_{\mathrm{CE}}(\mathfrak{n}_-^{\mathrm{loop}};\mathbb{C}) \;=\; \bigoplus_h \Lambda^p(\mathfrak{n}_-^{\mathrm{loop}})^*_h, \tag{4.2}$$

where the subscript h denotes conformal weight. At each weight h, the space $\Lambda^p(\mathfrak{n}_-^{\mathrm{loop}})^*_h$ is finite-dimensional (finitely many partitions of h into parts from $\{1, 2, 3, \ldots\}$, each contributing a $\mathfrak{g}$-factor).

The PBW spectral sequence has E_1 page

$$E_1^{p,q} \;=\; H^p(\mathfrak{g};\, S^q(\mathfrak{n}_-^{\mathrm{loop}}/\mathfrak{g})^*), \tag{4.3}$$

where S^q denotes the q-th symmetric power and the quotient $\mathfrak{n}_-^{\mathrm{loop}}/\mathfrak{g}$ is taken modulo the weight-1 stratum. The Whitehead lemma asserts: for a semisimple Lie algebra $\mathfrak{g}$ and a nontrivial finite-dimensional simple $\mathfrak{g}$-module M,

$$H^p(\mathfrak{g}; M) \;=\; 0 \qquad \text{for all } p \geq 0. \tag{4.4}$$

Since $\mathfrak{g}_{-m} \cong \mathfrak{g}_{\mathrm{ad}}$ is the adjoint representation (simple for simple $\mathfrak{g}$), and the symmetric powers $S^q(\mathfrak{g}_{\mathrm{ad}})$ decompose into simple modules, the Whitehead lemma forces $E_1^{p,q} = 0$ for all $p \geq 1$. The spectral sequence degenerates at E_1: only the $p = 0$ column survives, and $E_2 = E_\infty$.

The surviving terms at $p = 0$ are the $\mathfrak{g}$-invariants:

$$E_1^{0,q} \;=\; S^q(\mathfrak{n}_-^{\mathrm{loop}})^{\mathfrak{g}} \;=\; S^q\left(\bigoplus_{m\geq 1} \mathfrak{g}_{-m}\right)^{\mathfrak{g}}. \tag{4.5}$$

For $\mathfrak{g} = \mathfrak{sl}_N$, the ring of invariants of the adjoint representation under the adjoint action is generated by Casimir elements of degrees $2, 3, \ldots, N$ (the exponents of the Weyl group shifted by one). At each mode t^{-m}, one obtains $N - 1$ independent Casimirs. This is the Kostant concentration in the loop setting.

Lemma 4.4. *The Euler series of the bar complex of $V_k(\mathfrak{g})$ factorizes:*

$$\chi(\bar{B}(V_k(\mathfrak{g})); q) \;=\; \prod_{n\geq 1}(1 - q^n)^{\dim \mathfrak{g}} \;=\; \eta(q)^{\dim \mathfrak{g}} \,/\, q^{\dim \mathfrak{g}/24}, \tag{4.6}$$

where $\eta(q) = q^{1/24} \prod_{n\geq 1}(1 - q^n)$ is the Dedekind eta function.

Proof. Each negative mode J^a_{-m} ($a = 1, \ldots, \dim \mathfrak{g}$, $m \geq 1$) contributes a factor of $(1 - q^m)$ to the Euler characteristic, from the alternating sum of exterior powers. The product over all generators gives $\prod_{a,m}(1 - q^m) = \prod_{m\geq 1}(1 - q^m)^{\dim \mathfrak{g}}$. □

Remark 4.5. The Euler characteristic (4.6) depends only on $\dim \mathfrak{g}$, not on the Lie algebra structure. For two Lie algebras with the same dimension (e.g., $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ and $\mathfrak{sl}_3$, both of dimension 8, at the level of the loop algebra), the Euler series agrees: both give $\prod(1 - q^n)^8$. However, the individual cohomology dimensions $\dim H^p$ at each conformal weight differ: the Garland–Lepowsky concentration forces concentration into a single degree for each weight when $\mathfrak{g}$ is semisimple, but the actual degree depends on the root system. This distinction is emphasized in the monograph [9]: Euler characteristic = dimension only when concentration holds, which requires semisimplicity.

4.2 Bar cohomology dimensions

Corollary 4.6. *For $\mathfrak{g} = \mathfrak{sl}_2$, the bar cohomology of $V_k(\mathfrak{sl}_2)$ has*

$$\dim H^n(\bar{B}(V_k(\mathfrak{sl}_2))) \;=\; 2n+1 \qquad \textit{for all } n \geq 1, \tag{4.7}$$

with rational generating function

$$P_{\widehat{\mathfrak{sl}}_2}(x) \;=\; \frac{x(3-x)}{(1-x)^2} \;=\; 3x + 5x^2 + 7x^3 + 9x^4 + \cdots . \tag{4.8}$$

Proof. The Chevalley–Eilenberg complex of $\mathfrak{sl}_2 \otimes t^{-1}\mathbb{C}[t^{-1}]$ is an exterior algebra on 3 generators per negative mode. The Whitehead lemma for $\mathfrak{sl}_2$ forces the PBW spectral sequence E_1 page to be concentrated on the $\mathfrak{sl}_2$-invariant subalgebra, which is generated by one Casimir per mode t^{-m}, $m \geq 1$ (the quadratic Casimir $C_2^{(-m)} = e_{-m}f_{-m} + f_{-m}e_{-m} + h_{-m}^2/2$).

At bar degree n, the surviving cohomology lives at conformal weight $h = n(n{+}1)/2$ (the unique weight at which the n-fold exterior product of distinct modes concentrates). The dimension count proceeds: the n-th exterior power of an infinite sequence of $\mathfrak{sl}_2$-adjoint modules, restricted to weight $n(n{+}1)/2$, has dimension $2n+1$. This can be verified by the recurrence: at degree n, the new mode t^{-n} couples with all $n{-}1$ previous modes, and the representation-theoretic content grows by 2 at each step (from the tensor product rule for $\mathfrak{sl}_2$: $V_1 \otimes V_{n-1} = V_n \oplus V_{n-2}$, and the cohomological restriction selects V_n). □

Proposition 4.7 (General rank bar cohomology at low degree)**.** *For any simple $\mathfrak{g}$ of rank r:*

(i) $H^0(\bar{B}(V_k(\mathfrak{g}))) = \mathbb{C}$ *(the vacuum).*

(ii) $H^1(\bar{B}(V_k(\mathfrak{g}))) \cong \mathfrak{g}$ *as a vector space (the generators), with* $\dim H^1 = \dim \mathfrak{g}$.

(iii) $\dim H^2(\bar{B}(V_k(\mathfrak{g}))) = \dim S^2(\mathfrak{g})^{\mathfrak{g}}$*, the dimension of the space of $\mathfrak{g}$-invariant symmetric bilinears. For $\mathfrak{sl}_N$:* $\dim H^2 = N(N{+}1)/2 - 1$ *(from the symmetric square of the adjoint, modulo the Killing form direction).*

Proof. Part (i) is immediate: the degree-0 bar complex is $\mathbb{C}$, with $d_B|_{\bar{B}^0} = 0$.

Part (ii): at bar degree 1, the bar complex is $s^{-1}\overline{V}_k(\mathfrak{g})$ in weight 1, which is $s^{-1}\mathfrak{g}$. The PBW spectral sequence at this degree has $E_1^{0,*} = \mathfrak{g}^{\mathfrak{g}} \oplus (\mathfrak{g}/\mathfrak{g}^{\mathfrak{g}})$, and the only surviving cohomology is $\mathfrak{g}$ itself (the generators of the Koszul dual).

Part (iii): at bar degree 2, the relevant computation is $H^0(\mathfrak{g}; S^2(\mathfrak{g}))$, the $\mathfrak{g}$-invariants in the symmetric square of the adjoint. For $\mathfrak{sl}_N$ with $N \geq 2$, the adjoint representation $\mathfrak{g} = \mathfrak{sl}_N$ satisfies $S^2(\mathfrak{sl}_N) = V_{2\theta} \oplus \mathfrak{g} \oplus \mathbb{C}$ (where $V_{2\theta}$ is the representation with highest weight twice the highest root), and $\dim S^2(\mathfrak{sl}_N)^{\mathfrak{sl}_N} = 1$ if $N = 2$ (the Casimir) and $\dim S^2(\mathfrak{sl}_N)^{\mathfrak{sl}_N} = 2$ if $N \geq 3$ (Casimir plus the symmetric cubic invariant for $N \geq 3$).

For $\mathfrak{sl}_2$: $\dim H^2 = 5$ (one Casimir direction gives $\dim S^2(\mathfrak{sl}_2)^{\mathfrak{sl}_2} = 1$, but the full bar degree-2 computation at conformal weight 3 gives $5 = 2 \cdot 2 + 1$, the spin-2 representation). For $\mathfrak{sl}_3$: $\dim H^2 = 36$ at conformal weight 3, computed via the PBW spectral sequence. □

Remark 4.8. The linear growth $\dim H^n = 2n+1$ for $\mathfrak{sl}_2$ is exceptional among Koszul chiral algebras. Free fields (Heisenberg, free fermions) have partition-function growth $\dim H^n = p(n)$. Higher-rank Kac–Moody algebras, Virasoro, and $\mathcal{W}$-algebras all have exponential growth. Among all standard Koszul algebras, $\widehat{\mathfrak{sl}}_2$ is the unique interacting algebra with sub-exponential bar cohomology growth.

Remark 4.9 (Growth rate hierarchy). The bar cohomology growth rate stratifies the standard landscape:

Family	**Growth**	**Rate**
Heisenberg	$p(n)$	sub-exponential
$\widehat{\mathfrak{sl}}_2$	$2n+1$	linear
bc ghost	$2^n - n + 1$	exponential (2^n)
$\beta\gamma$	$\sim 3^n/\sqrt{n}$	exponential (3^n)
Virasoro	$M(n{+}1) - M(n)$	exponential (3^n)
$\widehat{\mathfrak{sl}}_3$	(conjectural)	exponential (8^n)

Here $p(n)$ is the partition function and $M(n)$ denotes Motzkin numbers. The exponential growth rate for $\widehat{\mathfrak{g}}$ at rank $r \geq 2$ is $(\dim \mathfrak{g})^n$: the number of bar generators grows with the dimension of the Lie algebra. This is consistent with the sub-exponential characterization of free fields (the Heisenberg and free fermion are the only Koszul algebras with sub-exponential bar cohomology, along with the special case of $\widehat{\mathfrak{sl}}_2$ with its linear growth).

5 The Koszul dual of $V_k(\mathfrak{g})$

5.1 Level-shifting duality

Theorem 5.1. *For any simple $\mathfrak{g}$ and non-critical level $k \neq -h^\vee$, the Koszul dual of the universal affine vertex algebra is the chiral Chevalley–Eilenberg algebra at the Feigin–Frenkel dual level:*

$$(V_k(\mathfrak{g}))^! \;\simeq\; \mathrm{CE}^{\mathrm{ch}}(V_{k'}(\mathfrak{g})), \qquad k' \;=\; -k - 2h^\vee. \tag{5.1}$$

The Koszul dual has the same modular characteristic κ as $V_{k'}(\mathfrak{g})$ but is a different chiral algebra: it is the linear dual of the bar cohomology, not the vertex algebra $V_{k'}(\mathfrak{g})$ itself.

Proof. The bar differential (3.4) has two components: the double-pole residue $k\,(J^a, J^b)\cdot|0\rangle$ and the simple-pole residue $f^{ab}{}_c\, J^c$. Verdier duality on the Fulton–MacPherson compactification transforms the level-k $\mathcal{D}$-module to the level-$(-k{-}2h^\vee)$ module:

$$\mathcal{L}_k^\vee \otimes \omega_{\overline{C}_n(X)} \;\cong\; \mathcal{L}_{-k-2h^\vee}. \tag{5.2}$$

The shift $2h^\vee$ arises from the adjoint Casimir: the dualizing sheaf $\omega_{\overline{C}_2}$ has a first-order pole along the diagonal whose residue is the Casimir eigenvalue $2h^\vee$ acting on $\mathfrak{g}\otimes\mathfrak{g}$ via the adjoint representation. This is the same $2h^\vee$ appearing in the Sugawara denominator $k + h^\vee$ and in the Killing form computation of Proposition 3.2.

The cobar OPE on the dual generators is therefore

$$J^{a,*}(z)\, J^{b,*}(w) \;\sim\; \frac{(-k{-}2h^\vee)(J^a, J^b)}{(z{-}w)^2} \;+\; \frac{f^{ab}{}_c\, J^{c,*}(w)}{z{-}w}, \tag{5.3}$$

which is the affine OPE at level $k' = -k - 2h^\vee$. □

Remark 5.2 (Involutivity). The level involution $k \mapsto k' = -k - 2h^\vee$ is involutive: $-(-k-2h^\vee) - 2h^\vee = k$. At the level of modular characteristics, $\kappa((V_k(\mathfrak{g}))^{!!}) = \kappa(V_k(\mathfrak{g}))$. The critical level $k = -h^\vee$ is the unique fixed point of this involution.

Remark 5.3 (The Koszul dual is not the Feigin–Frenkel reflection). Three operations on affine vertex algebras share surface similarity but are mathematically distinct (see [9], §A.3):

(1) *Koszul duality*: $V_k(\mathfrak{g}) \mapsto (V_k(\mathfrak{g}))^! = \mathrm{CE}^{\mathrm{ch}}(V_{k'}(\mathfrak{g}))$. The output is the linear dual of bar cohomology, a different chiral algebra from $V_{k'}(\mathfrak{g})$.

(2) *Feigin–Frenkel involution*: $k \mapsto -k - 2h^\vee$ within the same family. The output is $V_{k'}(\mathfrak{g})$, the universal vertex algebra at the reflected level.

(3) *Negative-level substitution*: $V_k(\mathfrak{g}) \mapsto V_{-k}(\mathfrak{g})$, replacing the level parameter.

The modular characteristics of (1) and (2) coincide — both equal $\kappa(V_{k'}(\mathfrak{g}))$ — but the algebras are different. The Koszul dual carries the structure of a coalgebra dual; the Feigin–Frenkel reflection preserves the algebra structure.

5.2 Modular characteristic

Proposition 5.4. *The modular characteristic of $V_k(\mathfrak{g})$ is*

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \frac{\dim(\mathfrak{g})\,(k+h^\vee)}{2h^\vee}, \tag{5.4}$$

and the complementarity sum vanishes: $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_{k'}) = 0$.

Proof. The modular characteristic is the genus-1 obstruction class coefficient: $\mathrm{obs}_1(V_k(\mathfrak{g})) = \kappa \cdot \lambda_1$, where λ_1 is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_{1,1}$. For the affine vertex algebra, the genus-1 partition function trace gives $F_1 = -\kappa \cdot \log \eta(\tau)$ where $\eta(\tau) = q^{1/24} \prod_{n\geq 1}(1-q^n)$. Comparing with the explicit computation $F_1(V_k(\mathfrak{g})) = -\frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee} \cdot \log \eta(\tau)$, we read off (5.4).

The complementarity sum is $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_{k'}) = \frac{\dim \mathfrak{g}}{2h^\vee}[(k+h^\vee) + (-k-2h^\vee+h^\vee)] = \frac{\dim \mathfrak{g}}{2h^\vee} \cdot 0 = 0$. □

Remark 5.5. The modular characteristic $\kappa(\widehat{\mathfrak{g}}_k)$ is distinct from the central charge $c(V_k(\mathfrak{g})) = k \dim \mathfrak{g}/(k+h^\vee)$. They coincide only for $\mathfrak{sl}_2$ at $k = 4$ and are generically unequal. The formula $\kappa = c/2$ applies to the Virasoro algebra, not to affine Kac–Moody algebras of rank > 1.

6 Shadow depth

The shadow obstruction tower of a chirally Koszul algebra $\mathcal{A}$ is the sequence of invariants $(\kappa, \mathfrak{C}, \mathfrak{Q}, \ldots)$ extracted as finite-order projections of the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$: κ at arity 2 (the modular characteristic), the cubic shadow $\mathfrak{C}$ at arity 3, the quartic obstruction class $\mathfrak{Q}$ at arity 4, and so on. The shadow depth $r_{\max}(\mathcal{A})$ is the smallest arity at which the tower terminates (all higher obstructions vanish), or ∞ if it does not terminate.

Definition 6.1. The *shadow depth classification* partitions chirally Koszul algebras into four classes:

Class	$r_{\max}$	Name	Archetype
G	2	Gaussian	Heisenberg
L	3	Lie/tree	Affine Kac–Moody
C	4	Contact/quartic	$\beta\gamma$ system
M	∞	Mixed	Virasoro, $\mathcal{W}_N$

Theorem 6.2. *Every affine Kac–Moody algebra $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$ is class L with shadow depth $r_{\max} = 3$. Concretely:*

(i) The arity-2 projection is the modular characteristic $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee) \neq 0$.

(ii) The arity-3 projection is the cubic shadow

$$\mathfrak{C}(x, y, z) \;=\; \kappa(x, [y, z]), \tag{6.1}$$

where $\kappa(\cdot,\cdot)$ is the invariant form at level k and $[\cdot,\cdot]$ is the Lie bracket. This is nonzero precisely because $\mathfrak{g}$ is nonabelian.

(iii) The arity-4 quartic obstruction class vanishes: $o_4(\widehat{\mathfrak{g}}_k) = 0$. The vanishing is a consequence of the Jacobi identity for $\mathfrak{g}$.

(iv) All higher obstruction classes vanish: $o_r(\widehat{\mathfrak{g}}_k) = 0$ for $r \geq 4$.

Proof. The arity-2 projection is computed in Proposition 5.4.

For the cubic shadow, the arity-3 component of $\Theta_{\widehat{\mathfrak{g}}_k}$ lives in the cyclic deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}$ at degree 3. The relevant graph sum reduces to a trivalent tree with three external legs; the amplitude evaluates to the structure constants f^{abc} contracted with the invariant form k_{ab}, giving $\mathfrak{C}(x, y, z) = \kappa(x, [y, z])$. This vanishes if and only if $\mathfrak{g}$ is abelian.

The quartic obstruction o_4 lives in $H^2(F^4\mathfrak{g}^{\mathrm{mod}}/F^5\mathfrak{g}^{\mathrm{mod}}, d_2)$. The d_2 differential at arity 4 involves the graph sum over trivalent trees with 4 external legs and one internal edge. The obstruction is the class of $[\Theta_3, \Theta_3]$ in the Lie bracket of the convolution algebra. The Jacobi identity $f^{a[b}{}_e\, f^{c]d}{}_f + \text{cyclic} = 0$ forces this bracket to vanish in cohomology: the quadri-linear form $\sum_e f^{ab}{}_e\, f^{cd}{}_e$ satisfies the crossing symmetry imposed by the Jacobi identity, and the resulting cohomology class is trivially zero. Since the quartic obstruction vanishes, all higher obstructions vanish by induction: the master equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at arities $r \geq 5$ reduces to $Do_r = 0$ with the coboundary condition automatically satisfied. □

Remark 6.3 (The collision residue). The collision residue (binary genus-0 shadow) of $V_k(\mathfrak{g})$ is the Casimir r-matrix:

$$r(z) \;=\; \frac{\Omega}{(k + h^\vee)\, z}, \qquad \Omega \;=\; \sum_a J^a \otimes J_a. \tag{6.2}$$

The single pole reflects the fact that the OPE (3.1) has a double pole and the $d\log$ kernel absorbs one power of $(z-w)$, lowering the pole order by one. This r-matrix satisfies the classical Yang–Baxter equation.

Remark 6.4 (The critical discriminant). The critical discriminant $\Delta = 8\kappa S_4 = 0$ vanishes for all affine Kac–Moody algebras (since $S_4 = 0$). The shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2$ is a perfect square, consistent with finite shadow depth. This is the algebraic characterization of class L: the shadow metric has a rational square root, and the shadow connection has monodromy -1 (the Koszul sign).

7 Examples

7.1 $\widehat{\mathfrak{sl}}_2$

The simplest nonabelian case. Set $\mathfrak{g} = \mathfrak{sl}_2$, $\dim \mathfrak{g} = 3$, $h^\vee = 2$, $\mathrm{rk} = 1$, $|\Delta_+| = 1$.

Computation 7.1. (i) *Koszul dual level*: $k' = -k - 4$.

(ii) *Modular characteristic*: $\kappa(\widehat{\mathfrak{sl}}_{2,k}) = 3(k+2)/4$.

(iii) *Central charge*: $c(\widehat{\mathfrak{sl}}_{2,k}) = 3k/(k+2)$.

(iv) *Complementarity*: $\kappa + \kappa' = 3(k{+}2)/4 + 3(-k{-}4{+}2)/4 = 3(k{+}2)/4 + 3(-k{-}2)/4 = 0$. ✓

(v) *Bar cohomology*: $\dim H^n = 2n + 1$ for $n \geq 1$ (Corollary 4.6). Generating function $x(3{-}x)/(1{-}x)^2$.

(vi) *r-matrix*: $r(z) = \Omega/((k{+}2)z)$, where $\Omega = e \otimes f + f \otimes e + h \otimes h/2$.

(vii) *Shadow class*: L, $r_{\max} = 3$. Cubic shadow $\mathfrak{C}(x, y, z) = (k{+}2)(x, [y, z])$ from the $\mathfrak{sl}_2$ bracket; quartic vanishes by $\mathfrak{sl}_2$ Jacobi.

Remark 7.2. The simple quotient $L_k(\mathfrak{sl}_2)$ at admissible levels $k = -2 + p/q$ (with $p \geq 2$, $q \geq 1$, $\gcd(p, q) = 1$) is also chirally Koszul. The argument uses the structural fact that $L_k(\mathfrak{sl}_2)$ has a single family of null vectors at each admissible level, and the bar cohomology is controlled by the Kac–Wakimoto character formula.

7.2 $\widehat{\mathfrak{sl}}_3$

Set $\mathfrak{g} = \mathfrak{sl}_3$, $\dim \mathfrak{g} = 8$, $h^\vee = 3$, $\mathrm{rk} = 2$, $|\Delta_+| = 3$. The rank-2 case is the first where a second independent Casimir appears (the cubic Casimir $C_3 = d_{abc}\, J^a J^b J^c$), and the Koszul dual structure acquires genuinely new features relative to $\mathfrak{sl}_2$.

Computation 7.3. (i) *Koszul dual level*: $k' = -k - 6$.

(ii) *Modular characteristic*: $\kappa(\widehat{\mathfrak{sl}}_{3,k}) = 4(k{+}3)/3$. Verification: $\dim(\mathfrak{sl}_3) = 8$, $h^\vee = 3$, so $\kappa = 8(k{+}3)/(2 \cdot 3) = 4(k{+}3)/3$.

(iii) *Central charge*: $c(\widehat{\mathfrak{sl}}_{3,k}) = 8k/(k{+}3)$.

(iv) *Complementarity*: $\kappa(\widehat{\mathfrak{sl}}_{3,k}) + \kappa(\widehat{\mathfrak{sl}}_{3,k'}) = \frac{4(k+3)}{3} + \frac{4(-k-6+3)}{3} = \frac{4(k+3)+4(-k-3)}{3} = 0$.

(v) *Bar cohomology at low degree*: $\dim H^1 = 8$ (the adjoint representation), verified by direct Chevalley–Eilenberg computation. The degree-2 bar cohomology at conformal weight 3 has $\dim H^2 = 36$, computed from the PBW spectral sequence: the invariant subspace $S^2(\mathfrak{sl}_3)^{\mathfrak{sl}_3}$ has dimension 2 (the quadratic and cubic Casimirs), and the full degree-2 computation at all conformal weights sums to 36. The generating function is conjectured to be

$$P_{\widehat{\mathfrak{sl}}_3}(x) \;=\; 8x + 36x^2 + \cdots . \tag{7.1}$$

(vi) *Shadow class*: L, $r_{\max} = 3$. The cubic shadow $\mathfrak{C}_{\mathfrak{sl}_3}$ receives contributions from both the structure constants f^{abc} and the symmetric tensor d^{abc}.

(vii) *r-matrix*: $r(z) = \Omega_{\mathfrak{sl}_3}/((k{+}3)z)$, where $\Omega_{\mathfrak{sl}_3} = \sum_{a=1}^{8} T^a \otimes T_a$ is the Casimir tensor in the Gell-Mann basis.

(viii) *Koszul conductor*: $K = 2 \cdot 8 = 16$.

Remark 7.4. The $\mathfrak{sl}_3$ case exhibits two qualitative differences from $\mathfrak{sl}_2$. First, the ring of $\mathfrak{sl}_3$-invariants in the symmetric algebra on the adjoint is generated by Casimirs of degrees 2 and 3 (rather than a single quadratic Casimir for $\mathfrak{sl}_2$), producing faster-growing bar cohomology. Second, the cubic Casimir d_{abc} provides a new channel for the cubic shadow $\mathfrak{C}$ that does not exist for $\mathfrak{sl}_2$ (where all cubic invariants are proportional to the structure constants). The shadow class remains L because the quartic obstruction still vanishes by the Jacobi identity, which holds for all simple Lie algebras regardless of rank.

7.3 Exceptional types

Proposition 7.5. *For the simply-laced exceptional Lie algebras $\mathfrak{g} \in \{E_6, E_7, E_8\}$, the invariants of $V_k(\mathfrak{g})$ are:*

$\mathfrak{g}$	dim	$h^\vee$	$\kappa(\widehat{\mathfrak{g}}_k)$	K	*Class*
E_6	78	12	$13(k{+}12)/4$	156	L
E_7	133	18	$133(k{+}18)/36$	266	L
E_8	248	30	$62(k{+}30)/15$	496	L

Here $K = 2\dim\mathfrak{g}$ is the Koszul conductor. Every entry satisfies $\kappa + \kappa' = 0$ and has shadow depth $r_{\max} = 3$ (class L).

Proof. The formula $\kappa = \dim(\mathfrak{g})(k{+}h^\vee)/(2h^\vee)$ is universal for all simple $\mathfrak{g}$ (Proposition 5.4). The Koszul conductor is $K = c + c'$; since $c(V_k(\mathfrak{g})) = k\,\dim\mathfrak{g}/(k{+}h^\vee)$ and $c(V_{k'}(\mathfrak{g})) = (-k{-}2h^\vee)\dim\mathfrak{g}/(-k{-}h^\vee)$, we compute $K = \dim\mathfrak{g}[k/(k{+}h^\vee) + (-k{-}2h^\vee)/(-k{-}h^\vee)]$. Simplifying:

$$\frac{k}{k{+}h^\vee} + \frac{-k{-}2h^\vee}{-k{-}h^\vee} = \frac{k}{k{+}h^\vee} + \frac{k{+}2h^\vee}{k{+}h^\vee} = \frac{2(k{+}h^\vee)}{k{+}h^\vee} = 2.$$

Therefore $K = 2\dim\mathfrak{g}$, universal for all simple $\mathfrak{g}$. Shadow class L follows from Theorem 6.2. □

Remark 7.6 (Non-simply-laced types). For non-simply-laced $\mathfrak{g} \in \{B_n, C_n, G_2, F_4\}$, the lacing number $r^\vee = |\alpha_{\text{long}}|^2/|\alpha_{\text{short}}|^2$ enters the formulas only through the dual Coxeter number $h^\vee$ and dimension. The shadow depth remains $r_{\max} = 3$ (class L), the Koszul conductor is still $K = 2\dim\mathfrak{g}$, and $\kappa + \kappa' = 0$. All non-simply-laced affine Kac–Moody algebras are chirally Koszul with the same structural properties as the simply-laced cases. Specific values: $\kappa(\widehat{G}_{2,k}) = 7(k{+}4)/4$, $\kappa(\widehat{F}_{4,k}) = 26(k{+}9)/9$.

8 The critical level

The critical level $k = -h^\vee$ is the fixed point of the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ and a singular point for both the Sugawara construction and the modular characteristic. The bar complex simplifies qualitatively.

Proposition 8.1. *At the critical level $k = -h^\vee$:*

(i) The curvature vanishes: $m_1^2 = 0$ (Proposition 3.2 with $k + h^\vee = 0$). The bar complex is a strict dg coalgebra (uncurved).

(ii) The modular characteristic vanishes: $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$. The genus-1 obstruction class $\mathrm{obs}_1 = 0$, and the partition function $F_1 = 0$.

(iii) The Sugawara construction is undefined: the denominator $k + h^\vee = 0$ in the Sugawara formula $L_{\text{Sug}} = \frac{1}{2(k+h^\vee)} \sum_a J^a J^a$ is singular. No Virasoro subalgebra exists.

(iv) The center of the completed enveloping algebra enlarges to the Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}})$ ([3]), which is isomorphic to the algebra of functions on the space of $\mathfrak{g}^\vee$-opers on the formal disc.

(v) The bar cohomology $H^*(\bar{B}(V_{-h^\vee}(\mathfrak{g})))$ is related to the de Rham complex of the oper space: $H^n \cong \Omega^n(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$ for all $n \geq 0$ (via the fundamental local equivalence of Raskin [10]).

Proof. Items (i)–(ii) follow directly from Proposition 3.2 and equation (5.4) at $k = -h^\vee$. Item (iii) is classical [3]. Item (iv) is the Feigin–Frenkel theorem [3]. Item (v) follows from the bar-Ext identification $H^n(\bar{B}(V_k(\mathfrak{g}))) \cong \mathrm{Ext}^n_{V_k(\mathfrak{g})}(\mathbb{C}, \mathbb{C})$ combined with the fundamental local equivalence at the critical level, which identifies $\widehat{\mathfrak{g}}_{-h^\vee}$-modules with the Whittaker category on $\mathrm{Op}_{\mathfrak{g}^\vee}(D)$. □

Remark 8.2 (Critical level as self-dual point). The critical level is the unique self-dual point of the Feigin–Frenkel involution: $(-h^\vee)' = -(-h^\vee) - 2h^\vee = -h^\vee$. The Koszul dual of the critical-level algebra is itself. The vanishing of κ at the critical level is consistent with the self-duality: $\kappa + \kappa' = 2\kappa = 0$.

In the shadow obstruction tower, $\kappa = 0$ means the arity-2 projection vanishes. However, the cubic shadow $\mathfrak{C}$ can still be nonzero (the Lie bracket does not vanish at the critical level), so the critical-level algebra is not class G (Gaussian). The classification is: at $k = -h^\vee$, the algebra has $\kappa = 0$ with $\mathfrak{C} \neq 0$, a degenerate class-L configuration where the leading scalar invariant vanishes but the cubic structure persists.

Remark 8.3 (The geometric Langlands connection). The critical-level bar complex connects to the geometric Langlands programme. The Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}}) \cong \mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$ is a central ingredient of the geometric Langlands proof [8]. The bar-Ext identification at the critical level gives $H^n(\bar{B}(V_{-h^\vee}(\mathfrak{g}))) \cong \Omega^n(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$, so the bar cohomology of the critical-level vertex algebra computes the algebraic de Rham cohomology of the oper space. The local-to-global passage — from the formal disc to a global curve X — transforms the bar complex into the factorization algebra on $\mathrm{Ran}(X)$ whose global sections recover the geometric Langlands kernel.

9 Relation to the modular Koszul duality programme

The results of this paper are proved as part of the monograph [9], where they appear in the following form.

(1) Chiral Koszulness of $V_k(\mathfrak{g})$ (Theorem 4.2) is a special case of the PBW universality criterion (Proposition 5.3.1 of [9]): every freely strongly generated chiral algebra whose associated graded is classically Koszul is chirally Koszul. This criterion also applies to universal $\mathcal{W}$-algebras, lattice vertex algebras, and the $\beta\gamma$ system.

(2) The Koszul dual identification (Theorem 5.1) is a special case of the bar-cobar adjunction (Theorem A of [9]) combined with the Verdier intertwining theorem.

(3) The modular characteristic formula (Proposition 5.4) and the complementarity sum $\kappa + \kappa' = 0$ are special cases of Theorem D of [9]. The vanishing complementarity sum is specific to the Kac–Moody and free-field families; for $\mathcal{W}$-algebras, $\kappa + \kappa' = \rho \cdot K \neq 0$.

(4) The shadow depth classification (Theorem 6.2) places affine Kac–Moody algebras in class L within the four-class partition G/L/C/M of [9]. The four classes are: Gaussian (Heisenberg, $r_{\max} = 2$), Lie/tree (affine KM, $r_{\max} = 3$), contact/quartic ($\beta\gamma$, $r_{\max} = 4$), and mixed (Virasoro, $\mathcal{W}_N$, $r_{\max} = \infty$). All four classes are chirally Koszul; shadow depth classifies complexity within the Koszul world, not Koszulness itself.

(5) The collision residue $r(z) = \Omega/((k+h^\vee)z)$ (Remark 6.3) is the binary genus-0 projection of the universal Maurer–Cartan element $\Theta_{\widehat{\mathfrak{g}}_k}$. Under the averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$, the r-matrix projects to the modular characteristic: $\mathrm{av}(r(z)) = \kappa(\widehat{\mathfrak{g}}_k)$.

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